PYTHON VS JAVA

OSCAR
SITGES

SINTAXIS

VS

```python
print("Hello World")
```

```python
def main():
    print("Hello World")

if __name__ == '__main__':
    main()
```

```java
public class JavaApplication1 {
    public static void main(String[] args) {
        System.out.println("Hello World");
    }
}
```

OPERADORES

- +, -, /, *, %...
- ==, <=, <...
- ...

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OPERADORES

VS

- +, -, /, *, %...
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** (POT)

// (DIV ENTE)

- +, -, /, *, %...
- ==, <=, <...
- ...

VARIABLES

VS

```python
def main():
    a = 3
    b = "Hola"


if __name__ == '__main__':
    main()
```

```java
public class JavaApplication1 {
    public static void main(String[] args) {
        System.out.println("Hello World");
        int a;
        String b;
    }

}
```

BUCLES

VS

```python
def main():
    for i in range(1, 11):
        print(i)

    a = 1
    while a < 10:
        print("¡Hola, mundo!")


if __name__ == '__main__':
    main()
```

```java
int a;
String b;
a = 0;

while(a <10){
    a++;
}

for (int i = 0; i < 10; i++) {

}
```

CONDICIONAL

```python
edad = 24
if edad < 18:
    print("Es usted menor de edad")
else:
    print("Es usted mayor de edad")
```

```java
int edad = 24;
if (edad < 18) {

    System.out.println(""
            + "Es usted menor de edad");
} else {
    System.out.println(""
            + "Es usted mayor de edad");
}
```

LISTAS Y DICCIONARIOS

VS

```python
lista = ['a', 'b', 'd', 'i', 'j']
print(lista[0])
```

```python
diccionario = {1: "A", 2: "B"}
print(diccionario[1])
```

```java
List ejemploLista = new ArrayList();
ejemploLista.add("a");
ejemploLista.add("b");
ejemploLista.add("c");
```

```java
Map<String, Integer> map =
        new HashMap<String, Integer>();

map.put("valor1", 15);
map.put("valor2", 20);
map.put("valor3", 1000);
map.put("valor4", 1500);
map.put("valor5", 2);
```

CONJUNTOS Y TUPLAS

VS

```python
s = {1, 2, 3, 4, 3}
print(list(s))
```

```python
tupla = 1, 2, 3
print(tupla)
```

```java
HashSet hset
        = new HashSet();
hset.add(1);
hset.add(2);
hset.add(3);
hset.add(4);
hset.add(3);
System.out.println(hset);
```

IMPORTAR JAVATUPLES

EJEMPLO Y CODIGO

```python
def fibonacci(n):
    if n < 2:
        return n
    fib_prev, fib = 1, 1
    for _ in range(2, n):
        fib_prev, fib = fib, fib + fib_prev
    return fib
```

```java
public static long fibonacci(int n) {
    if (n < 2) {
        return n;
    }
    long ans = 0;
    long n1 = 0;
    long n2 = 1;
    for (n--; n > 0; n--) {
        ans = n1 + n2;
        n1 = n2;
        n2 = ans;
    }
    return ans;
}
```

PYTHON VS JAVA

GRACIAS